



Lockheed Martin's ADAM (Credit: [www.geek.com](http://www.geek.com))

ate small, inexpensive drones loaded with weapons to threaten the enemy and allied forces on the ground," says Daniel Miller, chief engineer for high energy laser integration at Lockheed Martin Skunk Works. "Because of their size, these drones are difficult to see, hard to catch on the radar and hard to shoot with conventional weapons."

A Q-53 radar has been developed that can detect and track such drone swarms and provide data to the command and control centre. The US is also developing a 60kW system equipped with multiple fibre lasers to generate high-power weapon beams to destroy enemy drone swarms. This laser weapon system can fire again and again, essentially creating unlimited groups of 'bullets.'

Through internal research and development, Lockheed Martin Engineers have successfully developed a system called ADAM (Area Defense Anti Munitions). This system has successfully defeated small boats and rockets, while Advanced Test High Energy Asset (ATHENA) system has stopped drones and trucks in their tracks.

Researchers are also developing cyber solutions to defeat enemy drone swarm attacks. These solutions contain a multi-spectral sensor system that detects and characterises drone swarms. Using cyber electromagnetic activity, swarms are disabled so that life threat to native war fighters and citizens is avoided.

Today, the world is rocked by ter-

rorism and destructive nuclear wars. Can the world recover? Will drone swarms solve the problems threatening the world?

Unlike expensive combat jets, drone swarms are cost-effective and can detect enemy targets and strike them down with small weapon loads. So, global militaries are slowly choosing such swarms to step down anti-military efforts.

Drones can easily be 3D-printed in thousands and used as smart weapons to defeat the enemy with minimum casualties.

### Use of drone swarm technology in India

The global drone swarm market is huge, and India can cater to a good portion of that. While looking at this swarm technology, it needs an interactive remote control member. Speech technology is an effective and recent method to control swarms. Text-based communication is somewhat a difficult control option.

Another good control option is the use of secure mobile apps. India is researching to show that even network operators can be used for this purpose. Currently, India is trying to employ this new technology in its defence systems.

China has been focusing on this new field of drones by exploiting the gaps in US trade relationships. India has realised the importance of this new trend to maintain and enhance its defence operation capabilities rather late.

Traditional methods of acquiring new technology in the Indian armed forces involve the following:

- Defense Research and Development Organization (DRDO) for researching new technologies and developing new products
- Public sector undertakings like Bharat Electronics Ltd and Hindustan Aeronautics Ltd for manufacturing newly-developed defence products

This conventional method is quite lengthy with long procurement and development cycles.

For the first time, the Indian Air Force (IAF) is taking a different route to place a significant defence order (drone swarm) through an open contest. The move is being seen as a bid to counter the growing influence of startups. Winners of the contest will bag a defence contract for swarms worth millions of rupees.

Calling all drone manufacturing startups to join the contest, Anshuman Tripathi, a consultant working with IAF says, "This contest brings direct participation of startups and the MSME ecosystem to address the demands of the Indian defence industry." Three winners will get a prize money of one million rupees, followed by a co-production opportunity with Base Repair Depot (BRD). Winners will be announced on Kargil Vijay Day, that is, July 26, this year.

### Moving ahead

The most important challenge is the lack of awareness about the various benefits of drone swarms, in terms of economics, environment and performance. Quality, cost and delivery management is also a major area of challenge. Drone swarms should be highly synchronised with additional functional capabilities and reduced control features. This can ensure better performance going ahead.

Drone swarms are more than just a technology. These can become home for employment and economics, and can emerge as a business hub in the years to come. It is time to have a fresh perspective on this technology. Experts can apply technologies like AI, Big Data and the Internet of Things (IoT) to drone swarms to make these more effective. **EFY**